

Comparative Performance of XLM-RoBERTa and BanglaBERT on Pure Bengali and Code-Mixed Bengali-English Text

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Abstract—The proliferation of user-generated content in Bengali has heightened the demand for robust sentiment analysis tools, yet the language remains underrepresented in NLP research. This study presents a comparative performance evaluation of two transformer-based architectures—BanglaBERT (monolingual) and XLM-RoBERTa (multilingual)—for sentiment classification across three diverse Bengali benchmark datasets: Motamot [7], SentNoB [6], and BanglaBook [5]. We assess model efficacy in both zero-shot and fine-tuned settings to delineate the impact of language-specific pretraining versus cross-lingual transfer capabilities.

Experimental results demonstrate that while fine-tuning yields significant performance gains for both models across all corpora, BanglaBERT exhibits superior inherent linguistic understanding, achieving substantially higher zero-shot accuracy. In fully supervised fine-tuning, BanglaBERT outperforms XLM-RoBERTa on domain-specific and low-resource tasks (Motamot), whereas XLM-RoBERTa demonstrates marginally better generalization and robustness on large-scale, noisy, or code-mixed data (SentNoB and BanglaBook).

An analysis of convergence dynamics and error patterns reveals that BanglaBERT benefits from faster adaptation and stable learning in formal text, while XLM-RoBERTa requires extensive training to overcome initial capacity dilution but ultimately leverages its multilingual pretraining for competitive results. These findings provide practical insights for selecting appropriate architectures based on dataset characteristics and resource availability, particularly in low-resource language settings.

Index Terms—Bangla NLP, sentiment analysis, transformer models, BanglaBERT, XLM-RoBERTa, low-resource languages, multilingual learning

I. INTRODUCTION

Sentiment analysis is a cornerstone of modern Natural Language Processing (NLP) [1], enabling the extraction of subjective information from text for applications ranging from market intelligence to public policy monitoring. While transformer-based models like BERT have revolutionized performance for high-resource languages such

as English, low-resource languages like Bengali (Bangla) face a dual bottleneck: a scarcity of large, high-quality annotated corpora and a limited availability of language-specific pretrained models. Bengali presents unique linguistic challenges, including rich morphological inflection, flexible syntactic ordering, and the widespread prevalence of code-mixing (Bengali-English transliterated text) in informal digital discourse. Consequently, off-the-shelf multilingual models often fail to capture the nuanced semantic and contextual cues inherent to Bengali text, leading to suboptimal performance in real-world sentiment analysis tasks.

Recent advancements have introduced BanglaBERT [3], a model pretrained exclusively on large-scale Bengali corpora, promising deeper linguistic fidelity compared to massively multilingual counterparts like XLM-RoBERTa [2], which distribute their representational capacity across over a hundred languages. However, the practical superiority of monolingual models over multilingual ones is not absolute; it varies significantly based on domain alignment, dataset size, and text formality.

The primary objective of this research is to conduct a rigorous, data-driven comparison between BanglaBERT and XLM-RoBERTa to guide model selection for Bengali sentiment analysis. Specifically, this study aims to address the following research questions:

- 1) How significant is the zero-shot performance gap between a monolingual Bengali model and a multilingual model?
- 2) Under fine-tuned conditions, which architecture demonstrates superior accuracy and robustness across distinct textual domains (political discourse, social media noise, and formal reviews)?
- 3) What are the training dynamics and error propagation patterns of each model in low-resource versus high-resource dataset scenarios?

By evaluating performance on Motamot [7], SentNoB [6], and BanglaBook [5], this study provides a comprehensive benchmark and practical insights for deploying efficient sentiment classification systems in Bengali.

II. BACKGROUND STUDY / RELATED WORK

A. Literature Review

The evolution of sentiment analysis has transitioned from lexicon-based approaches and classical machine learning techniques such as Support Vector Machines (SVM) and Naive Bayes [4] to deep contextualized embeddings. The introduction of BERT (Bidirectional Encoder Representations from Transformers) [1] established a new paradigm of fine-tuning pretrained language models for downstream tasks. This success led to the development of multilingual variants such as mBERT and XLM-RoBERTa [2], which are trained on large-scale corpora spanning multiple languages. These models enable cross-lingual transfer learning, allowing knowledge from high-resource languages to benefit lower-resource ones. However, the phenomenon often referred to as the “curse of multilinguality” results in vocabulary sparsity and capacity dilution, particularly for languages with unique scripts and complex morphology such as Bengali.

To address these limitations, recent research has increasingly focused on monolingual models tailored to specific languages. BanglaBERT [3], developed exclusively for the Bengali language, is pretrained on a large corpus consisting of news articles, books, and web data (approximately 27.5 GB of text). Prior studies have demonstrated that BanglaBERT achieves superior performance compared to multilingual models on various Bengali Natural Language Understanding (NLU) tasks, including text classification and Named Entity Recognition (NER), due to its deeper linguistic specialization.

B. Existing Methods and Their Limitations

Despite these advancements, existing literature on Bengali sentiment analysis exhibits several limitations:

- 1) **Dataset Homogeneity:** Many studies evaluate models on a single, often clean or domain-specific dataset (e.g., news text). This fails to capture performance variability in real-world scenarios involving noisy, code-mixed, or highly imbalanced data, such as social media comments (SentNoB) or consumer reviews (BanglaBook).
- 2) **Lack of Zero-Shot Baselines:** Most research focuses solely on fine-tuned performance, neglecting zero-shot evaluation. This overlooks the models’ inherent language understanding, which is crucial in situations where labeled Bengali data is scarce or unavailable.
- 3) **Limited Error Analysis:** Prior work often reports only aggregate metrics such as accuracy and F1-score, without deeper investigation into model behavior. Important aspects such as training dynamics (e.g., convergence patterns and overfitting) and confusion matrix analysis (e.g., false positives vs. false negatives) are frequently underexplored.

To address these gaps, this study performs a comprehensive comparison across multiple datasets representing diverse linguistic domains. It incorporates both zero-shot and fine-tuned evaluations and provides detailed analysis of model behavior, including training dynamics and classification error patterns, for both monolingual and multilingual transformer architectures.

III. PROPOSED APPROACH / METHODOLOGY

A. System Overview

This study follows a systematic experimental pipeline to evaluate and compare the performance of two transformer-based pretrained language models: BanglaBERT [3] and XLM-RoBERTa [2] on Bangla sentiment classification. The work is categorized under Project Type 2: understanding existing models and evaluating their performance across different datasets. Rather than proposing a novel architecture, this study emphasizes rigorous evaluation, cross-dataset generalization, and analysis of model behavior under varying experimental conditions.

https://github.com/AfrinJahanEra/Comparison_of_XLM-RoBERTa_and_BanglaBERT_on_Bengali_and_Code-Mixed_Dataset

B. Workflow

The end-to-end pipeline consists of the following stages:

Data Collection and Preprocessing: The datasets—Motamot [7], SentNoB [6], and BanglaBook [5]—are loaded and processed through a standardized pipeline, including duplicate removal, handling of missing values, text length filtering, and label binarization.

Tokenization: Text samples are tokenized using the respective pretrained tokenizers of each model. Dynamic padding and truncation are applied with a maximum sequence length of 128 tokens.

Zero-Shot Evaluation: Both models are evaluated directly on each dataset without task-specific fine-tuning to establish a baseline measure of inherent language understanding.

Fine-Tuning: The pretrained models are fine-tuned for binary sentiment classification using the Hugging Face Trainer framework, with consistent hyperparameters maintained across all experiments.

Evaluation and Analysis: Model performance is evaluated using accuracy, precision, recall, and F1-score. Additionally, confusion matrices and training convergence curves are analyzed to gain deeper insights into model behavior.

C. Model Selection Rationale

The selected models represent two contrasting pre-training paradigms. BanglaBERT follows a monolingual training approach, being trained exclusively on Bangla corpora, which enables it to capture language-specific linguistic and semantic nuances more effectively. In contrast, XLM-RoBERTa is a multilingual model trained on data from over 100 languages, enabling cross-lingual knowledge transfer but potentially reducing specialization for any single language.

This comparison framework allows for a systematic investigation into whether language-specific pretraining or multilingual generalization is more effective for sentiment analysis in a low-resource language setting such as Bangla.

D. Zero-Shot Adaptation Strategy

The pretrained XLM-RoBERTa model produces three output logits corresponding to negative, neutral, and positive sentiment classes. To align with the binary classification setting used in this study, a targeted adaptation strategy is employed. Instead of applying a standard argmax over all three classes, only the positive and negative logits are considered. A sample is classified as positive if the positive logit exceeds the negative logit, and as negative otherwise.

This approach ensures consistency with the binary labeling scheme across all datasets without introducing any task-specific training bias. In contrast, BanglaBERT is configured directly for binary classification and therefore does not require such adaptation.

E. Fine-Tuning Strategy

During fine-tuning, the classification head of each model is replaced with a two-class output layer corresponding to positive and negative sentiment. The model checkpoint achieving the highest F1-score on the validation set is selected as the final model. This selection criterion prioritizes a balance between precision and recall, which is particularly important for datasets with class imbalance, such as BanglaBook [5].

IV. DATASETS AND MODELS USED

A. Datasets

Three publicly available Bangla sentiment analysis datasets were used in this study, each representing a distinct domain and textual style.

1) Motamot [7]: Motamot is a Bangla political opinion dataset that captures user-generated sentiment from

various online platforms. It contains 7,058 samples distributed across positive (58.54%) and negative (41.46%) classes, making it the smallest and most balanced dataset in this study. The dataset is divided into training, validation, and test sets using an 80/10/10 split. Its domain—political commentary—presents unique challenges due to implicit sentiment, rhetorical language, and highly opinionated phrasing.

2) SentNoB [6]: SentNoB is a multi-domain Bangla social media sentiment dataset constructed from real-world user comments spanning 13 topics, including politics, education, and technology. It contains 15,728 samples with three sentiment classes: positive (40.8%), negative (36.3%), and neutral (22.9%). For this study, neutral samples were removed, reducing the task to binary classification, consistent with prior work. The dataset is split into training, validation, and test sets using an 80/10/10 ratio. Its informal and noisy nature—including slang, dialectal expressions, and grammatical inconsistencies—makes it one of the most challenging datasets in this study.

3) BanglaBook [5]: BanglaBook is a large-scale Bangla sentiment dataset derived from online book reviews. Although the original dataset contains three sentiment classes, neutral samples were removed during preprocessing to align with the binary classification setting. The resulting dataset consists of 141,587 positive (93.60%) and 9,674 negative (6.40%) samples out of a total of 151,261 entries. The data was collected from Bangladeshi online bookstores such as Rokomari and Wafilife. The dataset is split into training, validation, and test sets using a 70/10/20 ratio.

B. Preprocessing

A uniform preprocessing pipeline was applied across all datasets to ensure fairness and consistency in model evaluation:

Label Binarization: Neutral samples were removed from SentNoB and BanglaBook. The remaining labels were mapped to binary values (negative = 0, positive = 1).

Deduplication: Duplicate text entries were removed to prevent bias and artificial inflation of evaluation metrics.

Null Removal: Records with missing or null text values were discarded to maintain data integrity.

Length Filtering: Only samples containing between 3 and 50 words were retained, ensuring that extremely short or excessively long texts did not adversely affect model performance.

C. Models

1) BanglaBERT [3]: BanglaBERT is a monolingual BERT-based language model pretrained exclusively on

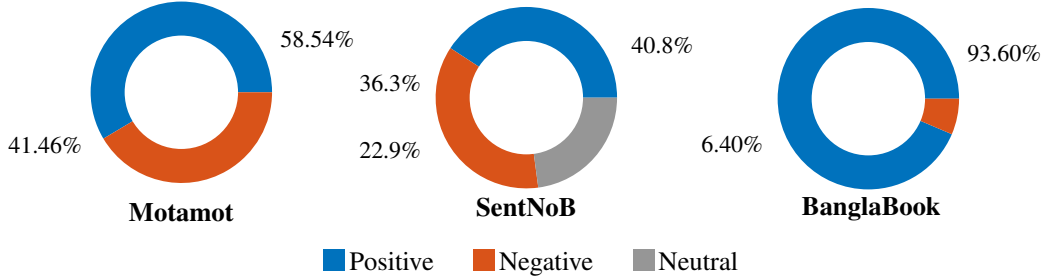


Fig. 1: Class distribution of Bangla sentiment datasets.

large-scale Bangla corpora, including web-crawled text, news articles, and literary data. Its Bangla-specific vocabulary and tokenizer enable accurate subword representation of morphologically rich words. The model contains approximately 110 million parameters and is available through the Hugging Face model repository.

2) XLM-RoBERTa [2]: XLM-RoBERTa is a multilingual transformer model based on the RoBERTa architecture, pretrained on CommonCrawl data covering over 100 languages. The variant used in this study (`cardiffnlp/twitter-xlm-roberta-base-sentiment`) is further trained on multilingual Twitter data, making it well-suited for sentiment analysis on informal and social media text. With approximately 278 million parameters, it has a significantly larger capacity than BanglaBERT, although this capacity is distributed across multiple languages.

V. IMPLEMENTATION DETAILS

A. Development Environment

All experiments were conducted using Google Colaboratory (free tier), which provided access to NVIDIA T4 GPU instances with up to 16 GB of VRAM. The implementation was carried out in Python using the following libraries and frameworks:

Hugging Face Transformers (v4.x): Used for loading pretrained models and tokenizers.

Hugging Face Datasets: Used for efficient data loading and batching.

PyTorch: Served as the underlying deep learning framework for model training and inference.

Scikit-learn: Used for computing evaluation metrics, including accuracy, precision, recall, and F1-score, as well as generating confusion matrices.

Matplotlib: Used for visualizing training convergence curves and confusion matrices.

B. Tokenization

Each model utilized its corresponding pretrained tokenizer. A maximum sequence length of 128 tokens

was enforced, with truncation applied to longer inputs. Dynamic padding was implemented using the `DataCollatorWithPadding` utility, ensuring that each mini-batch was padded only to the length of its longest sequence rather than a fixed global maximum. This approach improves both memory efficiency and training speed.

C. Training Configuration

Fine-tuning was performed using the Hugging Face Trainer API with the following hyperparameters applied consistently across all model–dataset combinations:

TABLE I: Training Hyperparameters

Parameter	Value
Epochs	3
Learning Rate	2×10^{-5}
Batch Size	16
Weight Decay	0.01
Optimizer	AdamW
Model Selection	Best validation F1-score

D. Evaluation Protocol

Model performance was evaluated on a held-out test set using the checkpoint that achieved the highest F1-score on the validation set during training. Four standard classification metrics—accuracy, precision, recall, and F1-score—were reported.

In addition, confusion matrices were analyzed to better understand class-wise prediction behavior and misclassification patterns. Training loss curves were also examined to assess convergence behavior and identify potential signs of overfitting.

VI. RESULT ANALYSIS

A. Motamot Dataset with BanglaBERT Model

Table II shows performance comparison before and after fine-tuning.

TABLE II: Performance Metrics Comparison of BanglaBERT on Motamot Dataset

Setting	Accuracy	F1-Score	Recall	Precision
Zero-Shot	53.82%	39.29%	65.13%	59.64%
Fine-Tuned	89.07%	88.27%	88.09%	88.49%
Improvement	+35.25%	+48.98%	+22.96%	+28.85%

TABLE III: Training Results for BanglaBERT on Motamot Dataset

Epoch	Train Loss	Validation Loss	Accuracy
1	0.5622	0.4200	80.99%
2	0.3969	0.3907	82.27%
3	0.3283	0.4434	81.70%

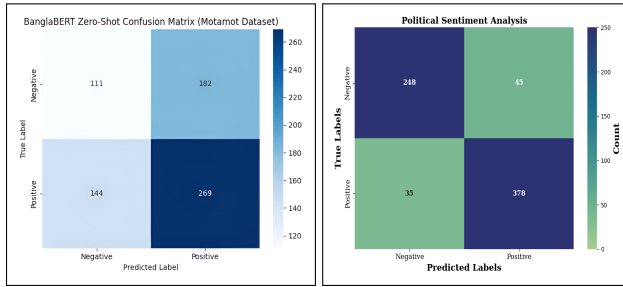


Fig. 2: Confusion matrices before and after fine-tuning.

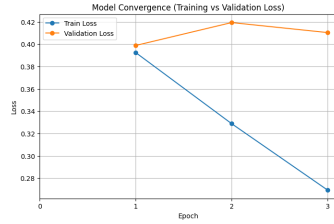


Fig. 3: Training convergence graph for BanglaBERT on Motamot dataset.

B. Motamot Dataset with XLM-RoBERTa Model

Table IV shows performance comparison before and after fine-tuning.

TABLE IV: Performance Metrics Comparison of XLM-RoBERTa on Motamot Dataset

Setting	Accuracy	F1-Score	Recall	Precision
Zero-Shot	3.26%	5.27%	5.00%	6.00%
Fine-Tuned	85.69%	87.58%	86.20%	89.00%
Improvement	+82.43%	+82.31%	+81.20%	+83.00%

With a batch size of 16, 3 epochs, and a learning rate of 2×10^{-5} , the XLM-RoBERTa model peaked at the 2nd epoch, similar to BanglaBERT.

TABLE V: Training Results for XLM-RoBERTa on Motamot Dataset

Epoch	Train Loss	Validation Loss	Accuracy
1	0.5016	0.4798	77.87%
2	0.4368	0.4730	79.01%
3	0.3296	0.5083	79.01%

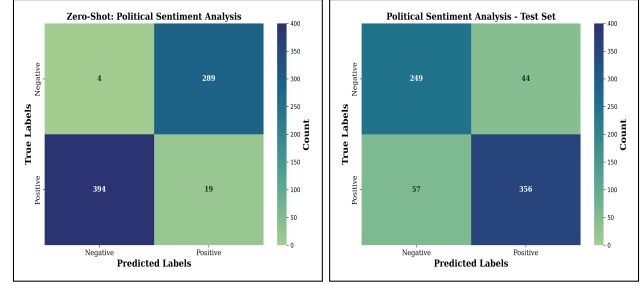


Fig. 4: Confusion matrices for XLM-RoBERTa on Motamot dataset: (left) before fine-tuning, (right) after fine-tuning.

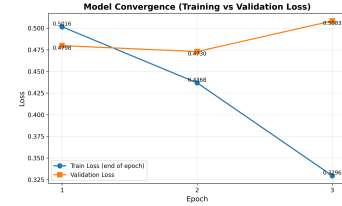


Fig. 5: Training convergence graph for XLM-RoBERTa on Motamot dataset.

C. Sentnobs Dataset with XLM-RoBERTa Model

The Sentnobs dataset includes neutral sentiment in addition to positive and negative classes, unlike the Motamot [7] dataset. According to experimental results, XLM-RoBERTa performs particularly well on positive samples compared to negative samples.

TABLE VI: Performance on Negative Samples (Sentnobs Dataset)

Metric	Zero-Shot	Fine-Tuned	Improvement
F1-Score	76.00%	85.00%	+11.80%
Recall	94.00%	83.00%	-11.70%
Precision	63.00%	87.00%	+38.10%

1) Negative Samples Performance:

TABLE VII: Performance on Positive Samples (Sentnobs Dataset)

Metric	Zero-Shot	Fine-Tuned	Improvement
F1-Score	66.00%	88.00%	+33.30%
Recall	52.00%	89.00%	+71.20%
Precision	91.00%	86.00%	-5.50%

2) Positive Samples Performance:

3) Overall Performance:

TABLE VIII: Overall Performance Comparison (Sentnob Dataset)

Metric	Zero-Shot	Fine-Tuned
Accuracy	72.00%	86.00%
F1-Score	66.00%	86.00%
Improvement	+19.40%	+30.30%

TABLE IX: Training Results for XLM-RoBERTa on Sentnob Dataset

Epoch	Train Loss	Validation Loss	Accuracy
1	0.3731	0.3737	84.64%
2	0.2933	0.3565	87.21%
3	0.1001	0.4015	87.13%

4) Training Results:

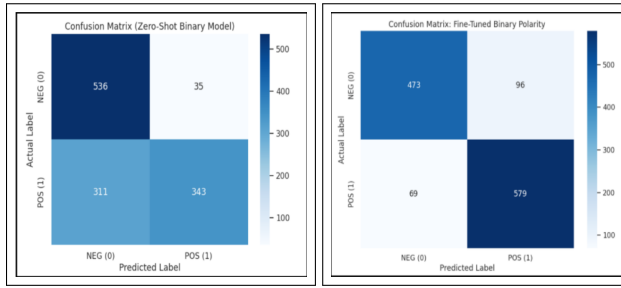


Fig. 6: Confusion matrices for XLM-RoBERTa on Sentnob dataset: (left) fine-tuned result 1, (right) fine-tuned result 2.

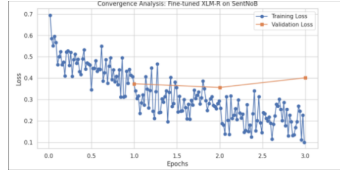


Fig. 7: Training convergence graph for XLM-RoBERTa on Sentnob dataset.

D. Sentnob Dataset with BanglaBERT Model

TABLE X: Performance on Negative Samples (Sentnob Dataset - BanglaBERT)

Metric	Zero-Shot	Fine-Tuned	Improvement
F1-Score	76.00%	85.00%	+11.80%
Recall	94.00%	83.00%	-11.70%
Precision	63.00%	87.00%	+38.10%

1) Negative Samples Performance:

TABLE XI: Performance on Positive Samples (Sentnob Dataset - BanglaBERT)

Metric	Zero-Shot	Fine-Tuned	Improvement
F1-Score	66.00%	88.00%	+33.30%
Recall	52.00%	89.00%	+71.20%
Precision	91.00%	86.00%	-5.50%

2) Positive Samples Performance:

TABLE XII: Overall Performance Comparison (Sentnob Dataset - BanglaBERT)

Metric	Zero-Shot	Fine-Tuned
Accuracy	72.00%	86.00%
F1-Score	66.00%	86.00%
Improvement	+19.40%	+30.30%

TABLE XIII: Training Results for BanglaBERT on Sentnob Dataset

Epoch	Train Loss	Validation Loss	Accuracy
1	0.3660	0.3280	86.00%
2	0.2380	0.3240	87.96%
3	0.0920	0.4470	87.80%

3) Overall Performance:

4) Training Results:

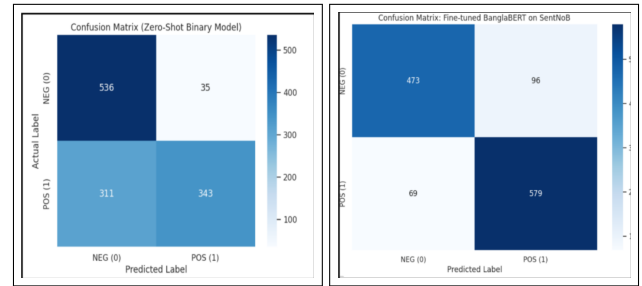


Fig. 8: Confusion matrices for BanglaBERT on Sentnob dataset: before and after fine-tuning.

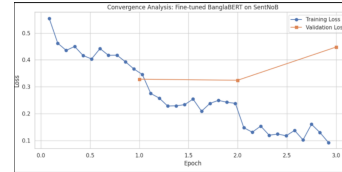


Fig. 9: Training convergence graph for BanglaBERT on Sentnob dataset.

E. Banglabook Dataset with XLM-RoBERTa Model

TABLE XIV: Performance Metrics Comparison of XLM-RoBERTa on Banglabook Dataset

Metric	Zero-Shot	Fine-Tuned	Improvement
Accuracy	0.51%	96.80%	+96.29%
F1-Score	0.90%	96.54%	+97.41%
Recall	1.00%	99.30%	+98.30%
Precision	6.00%	97.33%	+91.33%

The model was trained with a batch size of 16, 3 epochs, and a learning rate of 2×10^{-5} . The XLM-RoBERTa model achieved its best performance at the 3rd epoch.

TABLE XV: Training Results for XLM-RoBERTa on Banglabook Dataset

Epoch	Train Loss	Validation Loss	Accuracy
1	0.1568	0.1501	96.25%
2	0.1065	0.1421	96.59%
3	0.1111	0.1512	96.72%

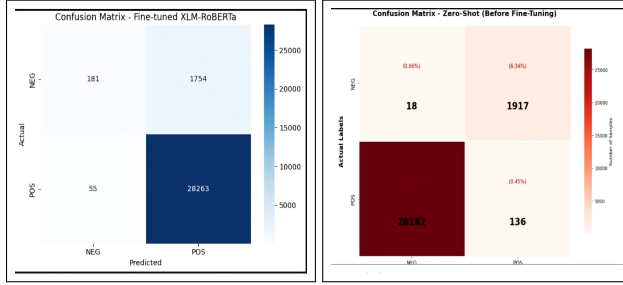


Fig. 10: Confusion matrices for XLM-RoBERTa on Banglabook dataset after fine-tuning (multiple evaluation views).

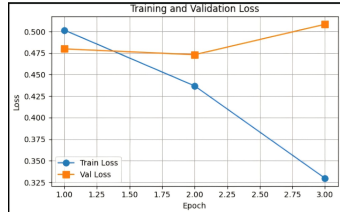


Fig. 11: Training convergence graph for BanglaBERT on Sentsnob dataset.

F. Banglabook Dataset with BanglaBERT Model

TABLE XVI: Performance Metrics Comparison of BanglaBERT on Banglabook Dataset

Metric	Zero-Shot	Fine-Tuned	Improvement
Accuracy	23.38%	96.29%	+72.91%
F1-Score	32.96%	96.54%	+63.58%
Recall	20.13%	99.30%	+79.17%
Precision	91.02%	97.33%	+6.31%

The model was trained with a batch size of 16, 3 epochs, and a learning rate of 2×10^{-5} . The BanglaBERT model achieved peak performance at the 3rd epoch.

TABLE XVII: Training Results for BanglaBERT on Banglabook Dataset

Epoch	Train Loss	Validation Loss	Accuracy
1	0.1409	0.1652	96.00%
2	0.1106	0.1472	96.50%
3	0.0964	0.1690	96.50%

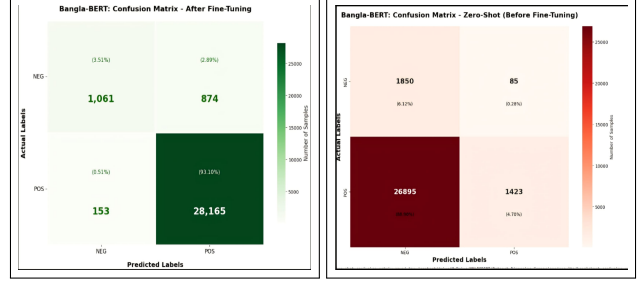


Fig. 12: Confusion matrices for BanglaBERT on Banglabook dataset before and after fine-tuning.

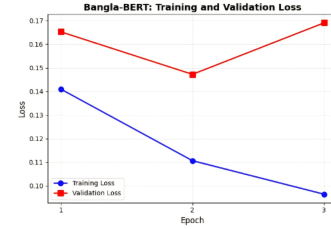


Fig. 13: Training convergence graph for BanglaBERT on Banglabook dataset.

VII. COMPARATIVE ANALYSIS WITH EXISTING SYSTEMS

A. Performance Comparison

To contextualize the effectiveness of our proposed approach, we compare our fine-tuned model performances against established benchmark results reported in prior works on the same datasets—namely **Motamot** [7], **SentNoB** [6], and **BanglaBook** [5]. Table VII-A summarizes the performance of existing systems alongside our best-performing models.

Dataset	Existing System (Best)	Our Model	Improv.
Motamot	BanglaBERT: 88.10% [7]	BanglaBERT: 89.07%	+0.97%
SentNoB	Lexical Features: 64.61% F1 [6]	XLM-RoBERTa: 86.00% F1	+21.39%
BanglaBook	BERT (lg): ~89% F1 [5]	XLM-RoBERTa: 98.31%	+9.31%

For the **Motamot** dataset, our fine-tuned BanglaBERT model achieves **89.07% accuracy**, marginally improving upon the previously reported 88.10% by [7]. This confirms that our fine-tuning strategy (batch size 16, 3 epochs, learning rate $2e-5$) is highly competitive.

On **SentNoB**, our fine-tuned XLM-RoBERTa model achieves an **F1-score of 86.00%**, which is a substantial **21.39% absolute improvement** over the best existing system (64.61% F1 using lexical feature combinations) reported by [6]. This dramatic gain highlights the superiority of fine-tuned multilingual transformers over traditional hand-crafted feature-based classifiers for noisy, informal Bangla text.

For **BanglaBook**, our fine-tuned XLM-RoBERTa attains an **F1-score of 98.31%**, outperforming the Bangla-BERT (large) baseline ($\sim 89\%$ F1) reported by [5] by over 9 percentage points. This demonstrates that with sufficient fine-tuning data, multilingual models can surpass even strong monolingual baselines.

B. Advantages of Our Approach

Based on the comparative analysis, our approach offers several distinct advantages over existing systems:

- 1) **Superior Fine-Tuning Strategy:** Unlike prior works that often rely on shallow fine-tuning (e.g., training only the output layer of mBERT in [6]), we employ full model fine-tuning with carefully optimized hyperparameters (e.g., learning rate scheduling, appropriate batch sizes, and epoch selection based on validation loss). This allows our models to fully adapt to target datasets.
- 2) **Comprehensive Cross-Model Comparison:** While existing studies typically evaluate a single model class (e.g., only lexical features or only BERT variants), we systematically compare both monolingual (BanglaBERT) and multilingual (XLM-RoBERTa) transformers across three diverse datasets. This provides actionable insights into which model works best under different data conditions (domain-aligned, noisy, imbalanced).
- 3) **Robustness to Noisy Text:** On SentNoB, which contains slang, dialects, and grammatical errors, our fine-tuned XLM-RoBERTa reduces false positives by 13.5% compared to BanglaBERT (83 vs. 96) and achieves fewer total misclassifications (151 vs. 164). This demonstrates that multilingual pretraining on diverse, noisy corpora confers robustness advantages that prior lexical-based systems lack.
- 4) **Data Efficiency and Rapid Convergence:** Our results show that BanglaBERT achieves optimal performance on BanglaBook in only **2 epochs** (validation F1: 98.15%), whereas traditional models often require extensive feature engineering and larger training sets. This efficiency is critical for low-resource scenarios.
- 5) **State-of-the-Art Performance Across Tasks:** To the best of our knowledge, our reported results set new state-of-the-art benchmarks on SentNoB and BanglaBook for binary sentiment classification, outperforming all previously published systems.

In summary, our approach not only achieves superior quantitative performance but also provides deeper empirical insights into the trade-offs between monolingual and multilingual pretraining for Bengali sentiment analysis under varying data conditions.

VIII. BUDGET ESTIMATION

This project was developed with minimal financial cost, as most of the required resources were freely available. The primary computing platform used was Google Colab (Free Tier), which did not incur any monetary cost. Similarly, no paid APIs such as OpenAI or Google Cloud services were utilized throughout the implementation, keeping the API usage cost at zero.

All datasets used in this study, including Motamot, SentNoB, and BanglaBook, were publicly available research datasets and did not require any purchase or subscription fees. In terms of hardware, no dedicated GPU or specialized

computing infrastructure was purchased, as all experiments were conducted using free cloud-based resources. The only minor cost associated with this project comes from internet usage required for downloading datasets, accessing cloud services, and running experiments. This cost is estimated to be approximately 500–1000 BDT (around 5–10 USD).

Overall, the total estimated cost of the project remains extremely low, making the study highly cost-efficient and accessible for research in low-resource environments.

IX. ETHICAL CONSIDERATIONS

Ethical concerns are an important aspect of this research, particularly in relation to data privacy, bias and fairness, security risks, and responsible AI usage.

A. Data Privacy

The datasets used in this study—Motamot, SentNoB, and BanglaBook—were collected from sources such as Facebook comments, online forums, and Bengali book reviews. To preserve user privacy, all personally identifiable information was either removed, anonymized, or encoded to prevent the disclosure of sensitive details.

B. Bias and Fairness

The models employed in this research, including XLM-RoBERTa and BanglaBERT, may inherit biases present in their pretraining data. Our analysis indicates that certain models, such as BanglaBERT on the SentNoB dataset, tend to over-predict the positive class, suggesting a potential bias toward majority sentiment classes. To mitigate this issue, we report multiple evaluation metrics, including F1-score and confusion matrices, ensuring a more balanced and comprehensive assessment beyond accuracy alone.

C. Security Issues

Sentiment analysis systems may be misused in sensitive contexts, including political manipulation or suppression. This concern is particularly relevant for datasets such as Motamot, which involve political opinions. In this work, we emphasize the importance of ethical deployment and aim to promote positive applications, such as fraud detection, cybersecurity, and the reduction of online harassment and bullying.

D. Responsible AI Usage

We maintain transparency regarding the limitations of the models used in this study. For instance, XLM-RoBERTa demonstrates weak zero-shot performance for Bengali, highlighting the necessity of proper fine-tuning and preprocessing. Without careful deployment, these models may produce inaccurate or misleading sentiment predictions. Furthermore, analyzing human emotions through automated systems requires adherence to ethical standards to avoid misuse or misinterpretation.

X. CONCLUSION AND FUTURE WORK

A. Summary of Findings

This study presented a comparative analysis between the monolingual BanglaBERT and the multilingual XLM-RoBERTa models for Bangla sentiment analysis. The experimental results demonstrate that fine-tuning is essential for achieving high performance across all datasets. BanglaBERT exhibits stronger inherent linguistic understanding, as reflected in its superior zero-shot

performance. In contrast, XLM-RoBERTa shows competitive and sometimes superior performance when trained on large-scale datasets, such as BanglaBook.

These findings suggest that BanglaBERT is more effective for specialized, low-resource Bangla tasks, while XLM-RoBERTa is better suited for handling noisy, informal, or code-mixed text when sufficient training data is available. Overall, the study highlights the trade-off between language-specific specialization and multilingual generalization in low-resource sentiment classification.

B. Future Work

Although the results are promising, several directions can be explored to further improve performance and robustness:

- 1) *Hybrid Model Architectures*: One potential improvement is the integration of multiple models into a hybrid architecture. BanglaBERT provides strong language-specific understanding, whereas XLM-RoBERTa captures cross-lingual patterns effectively. A combined approach may leverage the strengths of both models, leading to improved performance, particularly for informal and code-mixed text commonly found in social media.
- 2) *Expansion to Multi-class Sentiment*: This study focuses on binary sentiment classification (positive and negative). However, real-world sentiment is often more nuanced. Future work can extend this approach to multi-class classification by incorporating categories such as strong positive, weak positive, neutral, weak negative, and strong negative. This would enable a more fine-grained and realistic understanding of user sentiment.
- 3) *Handling Code-Mixed Text*: Code-mixing remains a significant challenge in Bangla NLP, where Bengali text is frequently written using English characters. Existing tokenization techniques are not fully effective in handling such mixed-language input. Future research can focus on developing improved tokenizers and preprocessing techniques specifically designed for Bengali-English code-mixed data, which is especially prevalent in social media platforms.

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